

Week 2 Discussion

Example Share

The following is an example of a top-scoring share. It is not the only way to earn full marks. A concise, well-sourced share with the three questions asked honestly and a concrete call can score just as well. This example illustrates the expected structure and depth of reasoning, not a length requirement.

Example: Gasoline Prices and How Much We Buy

Source: U.S. Energy Information Administration, [U.S. Regular All Formulations Gas Price](#), monthly, via FRED; and EIA, [U.S. Product Supplied of Finished Motor Gasoline](#), monthly.

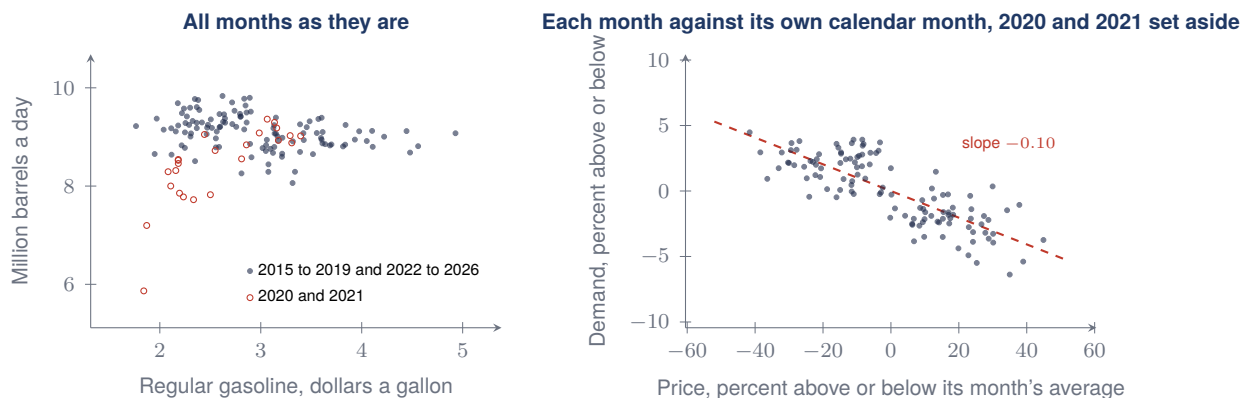
My Understanding

A line is an arrow only if changing X changes demand with everything else held still. Three things fake an arrow: a common shock that moves both, an arrow that runs the other way, and cases you cannot see. Behind a line can sit a log-log curve, where the slope is a percent-for-percent elasticity, or a U-shape, where the slope changes sign at a turning point.

Summary

From January 2015 to July 2026, regular gasoline averaged between \$1.76 and \$4.93 a gallon, and the country used between 5.9 and 9.8 million barrels a day. Plotted against each other, the 139 months make a cloud with no slope: $r = +0.08$, and a log-log line says a 1% higher price goes with 0.04% more gasoline. June 2019: \$2.72 and 9.70 million barrels a day. April 2020: \$1.84 and 5.87. June 2022: \$4.93 and 9.08.

No Line, Then a Line: U.S. Gasoline Price and Demand



Monthly data, January 2015 to July 2026. Left: price against barrels per day. Right: the same months, each measured against the average of its own calendar month, in log points, with 2020 and 2021 left out. The dashed line is the fitted log-log slope.

Reading the Shape

Start with the three questions. **Common shock:** two of them. The summer driving season raises both price and demand every year; in these data July demand runs about 10% above January and June prices about 20% above January. And the 2020 lockdown cut both at once, the cheapest gasoline in a decade and the least driving. Both shocks push the cloud toward a line that slopes up. **Reverse arrow:** within a summer, more driving pulls the price up, so part of any line runs from demand to price. **Missing cases:** none that I can find, though a national monthly total hides the drivers who cut a trip and the ones who did not.

Hold the two shocks still and the arrow appears. Compare each month with the average of its own calendar month, drop 2020 and 2021, and the log-log slope across 115 months is -0.10 with $r = -0.80$. A 10% higher price goes with about 1% less gasoline. That is a log-log reading, not a straight line in levels: it says the same percent at \$2 and at \$5. It also sits close to what researchers find for the short run, about -0.03 to -0.08 in the 2000s (Hughes, Knittel & Sperling, 2008). June 2022 is the test: price 81% above June 2019, demand 6% below.

No U-shape here. The dots never turn, and a range from \$1.76 to \$4.93 is wide enough that a turning point would have shown.

My Call

If I planned volumes for a chain of gas stations, I would forecast with the season and the economy, not with the price. A 10% price move shifts the country's demand by about 1% in a month; the summer alone shifts it by 10%. I would not treat -0.10 as my lever either. The elasticity at one station is far larger than the national one, because a ten-cent cut pulls drivers from the station across the street, so a price decision needs station-level data. I would revisit -0.10 if two or three years showed demand falling faster when prices rose, which could mean electric vehicles have changed who is left buying gasoline.

Where the Three Questions Fall Short

Elasticity depends on the window. Short-run elasticities are small because cars and commutes are fixed. Long-run ones are larger because people change cars and homes. One number is never the whole answer.

Which X ? The national price and the price at one station are different levers with different elasticities. The data must match the decision.

Holding the rest still is the whole game. Comparing a month with its own calendar month is a crude way to hold the season still, and dropping two years is a crude way to drop a lockdown. Researchers hold many things still at once, in a regression with several X 's. That is where the course goes next.

Going Deeper (Optional)

Seeing and doing. Pearl and Mackenzie (2018) call the two questions in this reading *seeing* and *doing*. Data can only show what happened when X moved for its own reasons. A decision moves X on purpose. Bridging the two is the subject of causal inference.

Spurious lines. Vigen (2015) collects dozens of pairs that move together with no arrow between them, such as cheese consumption and deaths by bedsheet entanglement. Most are

common shocks or common trends; a few are luck. A line with no story behind it deserves no decision.

Survivors. In the Second World War the statistician Abraham Wald was asked where to armor bombers, given the bullet holes on the planes that came back. He argued for armoring the places with no holes, because the planes hit there did not return (Mangel & Samaniego, 1984). Sales data are the planes that came back.

Testing a U-shape. A squared term with the right sign is not enough. The accepted test asks the slope to be positive at the low end of the data, negative at the high end, and the turning point to sit inside the range with a confidence interval that stays inside it (Lind & Mehlum, 2010; Haans, Pieters & He, 2016). The bike data pass: +479 rides per degree at the coldest day in the data, −111 at the hottest, and a peak at 29 degrees inside a range that runs from 2 to 35. This is the test used in management research, including your lecturer's own.

Elasticity as a tool. Log-log demand curves are the workhorse of pricing because the elasticity is the same number at every price. For gasoline, the shift from about −0.3 in the late 1970s to under −0.1 in the 2000s is itself a finding about how habits and car fleets change (Hughes, Knittel & Sperling, 2008).

Data, for the Discussion and this Example

Freddie Mac. [30-Year Fixed Rate Mortgage Average in the United States](#), weekly, via FRED.

Hass Avocado Board. [Retail volume and price data](#), weekly scan data, Total U.S., 2015 to March 2018, via the public Kaggle and OpenML mirror of the May 2018 file.

U.S. Census Bureau and U.S. Department of Housing and Urban Development. [New One Family Houses Sold](#), monthly, via FRED.

U.S. Energy Information Administration. [U.S. Regular All Formulations Gas Price](#) via FRED, and [U.S. Product Supplied of Finished Motor Gasoline](#), monthly.

UCI Machine Learning Repository. [Bike Sharing Dataset](#), daily file, Capital Bikeshare, Washington, DC, 2011 to 2012.

Wikimedia Foundation. [Pageviews API](#), monthly user views of the English Wikipedia articles *Air conditioning* and *Barbecue*.

References, for the Discussion and this Example

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Fanaee-T, H., & Gama, J. (2014). Event labeling combining ensemble detectors and background knowledge. *Progress in Artificial Intelligence*, 2(2–3), 113–127.

Haans, R. F. J., Pieters, C., & He, Z.-L. (2016). Thinking about U: Theorizing and testing U- and inverted U-shaped relationships in strategy research. *Strategic Management Journal*, 37(7), 1177–1195.

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Pearl, J., & Mackenzie, D. (2018). *The Book of Why: The New Science of Cause and Effect*. Basic Books.

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